

OpesCare — Complete PRD, Architecture & Module Map

Living document. Last updated: 2026-05-30. Update this file when modules ship or scope changes.

Source of truth derived from exhaustive codebase exploration of every route, controller, module, model, and config file.

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1. Product Vision

OpesCare is a **national-grade digital health platform** designed for Africa. It connects patients, clinicians, facilities, insurers, pharmacies, laboratories, NGOs, and government health authorities on a single interoperable infrastructure.

One-sentence pitch:

OpesCare gives every patient a portable, verifiable Health ID and gives every care provider — from a rural clinic to a national hospital — the digital tools to use it.

Core bets:

- Health ID as the anchor (one ID per patient, lifetime, shareable with consent)
- Consent-first data sharing (patient controls who sees what)
- Interoperability by default (FHIR R4, HL7, DHIS2 built in, not bolted on)
- Works offline and in low-connectivity (OpesCare Lite, offline sync)
- Multi-tenant SaaS (one platform, many facilities, one national registry)

- Multi-channel communications (Email, WhatsApp, SMS, Push, Voice)

2. Problem Statement

Problem	Who it affects	Current workaround
No portable patient identity across facilities	Patients	Paper cards, repeat registration
No patient consent layer for data sharing	Patients + Regulators	None — data shared without consent
Clinical data siloed per facility	Providers	Repeat tests, lost records
Insurance claims manual and slow	Insurers + Providers	Paper/email-based
Public health reporting fragmented	MoH / NGOs	Spreadsheets to DHIS2
Pharmacies can't verify prescriptions	Pharmacists	Phone calls
Labs can't deliver results digitally	Labs + Patients	Paper printouts
Referrals lost between facilities	Providers	Phone/WhatsApp
Rural facilities have no digital tools	Front-line workers	Paper registers
Clinical alerts not automated	Doctors	Manual chart review
Staff scheduling done manually	HR managers	Spreadsheets

3. Target Users & Personas

B2C — Patients & Families

Persona	Roles	Key Need
Patient	patient	See my records, control who accesses them
Guardian	guardian	Manage child's health ID and consent
Caregiver	caregiver	Temporary care delegation
Emergency Contact	emergency_contact	Access critical data in a crisis

B2B — Healthcare Providers

Persona	Roles	Key Need
Doctor / Specialist	doctor , specialist , consultant , resident , visiting_doctor	See full patient history, order labs/Rx
Nurse / Midwife	nurse , ward_nurse , midwife , nurse_supervisor	Triage, vitals, ward rounds, discharge
Triage Nurse	triage_nurse	Emergency triage scoring
Receptionist	receptionist , front_desk , appointment_coordinator	Queue, appointments, check-in

Doctor / Specialist	doctor , specialist , consultant , resident , visiting_doctor	See full patient history, order labs/Rx
Lab Technician	labtech , lab_scientist , lab_manager , lab_validator , sample_collection	Receive orders, enter results
Pharmacist	pharmacist , pharmacy_manager , pharmacy_technician , medicine_stock , dispensing_officer	Verify Rx, dispense, track inventory
Billing/Cashier	cashier , billing_officer , finance_manager , refund_approver , wallet_ops	Invoices, payments, reconciliation
Data Steward	data_steward , data_import_officer , data_quality_reviewer	Data imports, quality, correction
Queue Manager	queue_manager , records_officer	Queue and patient flow

B2B — Facility Administrators

Persona	Roles	Key Need
Facility Admin	facility_admin , clinic_admin , hospital_admin , facility_ceo , branch_admin	Configure facility, manage staff
Department Manager	department_manager	Department-level oversight
Privacy Officer	privacy_officer , data_protection_officer	Audit logs, consent reports, complaints
Security Officer	security_officer	Incidents, breach reports, access reviews
Finance Manager	finance_manager , finance	Revenue reports, billing oversight
Compliance Officer	compliance_officer , audit_reviewer	Audit exports, access reviews

B2B — Platform Admins

Persona	Roles
Platform Admin	platform_admin , super_admin , product_admin , system_admin
Legal / Country	legal_admin , country_admin , regional_admin
Support	support_agent , support_manager , customer_success , implementation_lead
Partner	partner_admin , partner_reviewer , partner_compliance , partner_technical
Academy	academy_admin

B2B — Insurance

Persona	Roles
Insurance Admin	insurance_admin
Claims Officer	insurance_claims
Preauth Reviewer	insurance_preauth
Finance	insurance_finance

Insurance Admin	insurance_admin
Reviewer	insurance_reviewer

B2G — Government & NGO

Persona	Roles
NGO Program Manager	health_program_manager, ngo_admin
Outreach Worker	outreach_team, mobile_clinic_team
Country/Regional Admin	country_admin, regional_admin

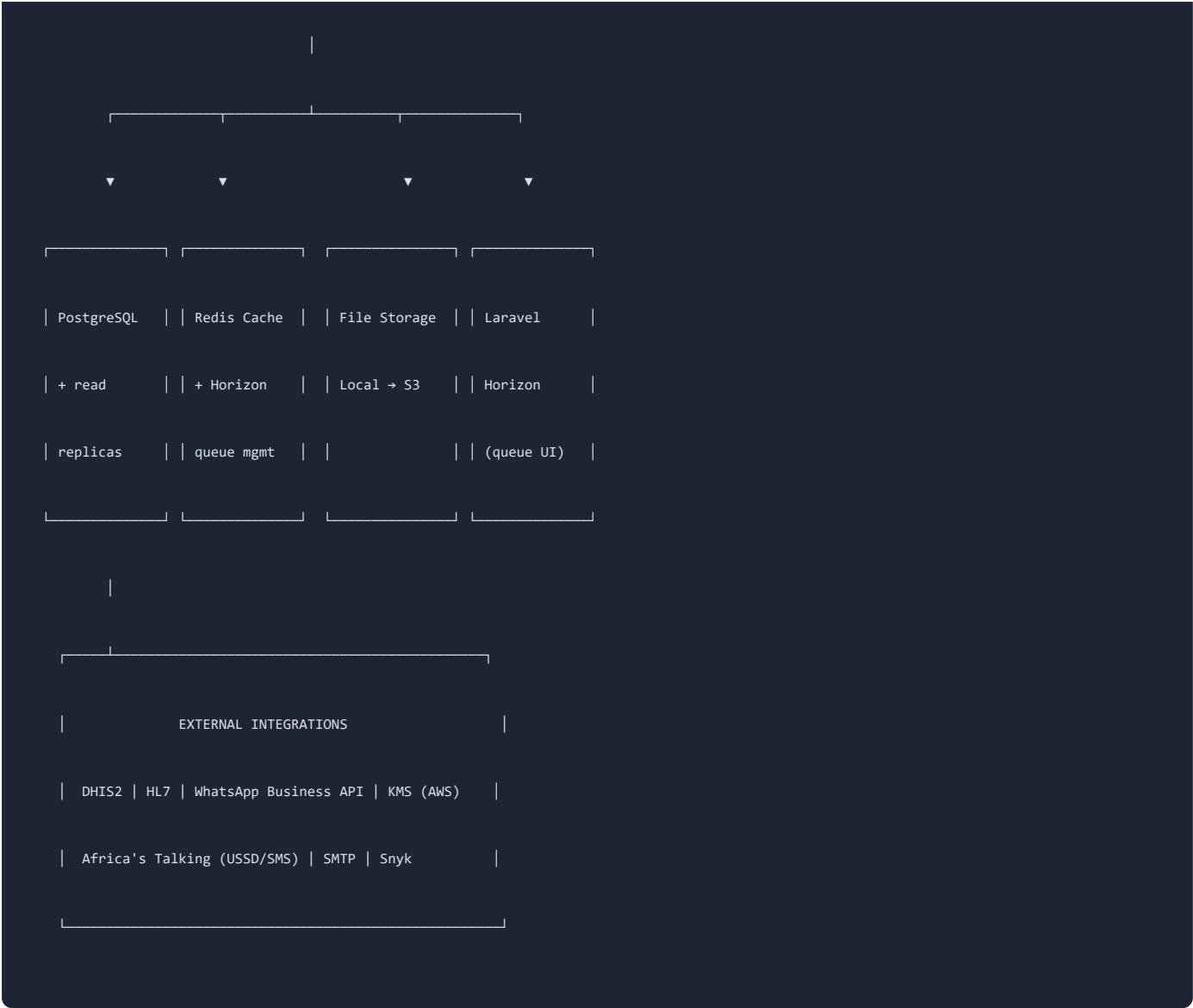
Developer / Partner / System

Persona	Roles
Developer	developer, developer_org_admin, api_partner, api_technical
Webhook Manager	webhook_manager
Sandbox Dev	sandbox_developer
Bridge Agent	System auth via VerifyBridgeAgent middleware

4. System Architecture







Request Flow — Web Portal

Browser → /portals/{portal}/{page}

|

▼

ForceHttps → AddSecurityHeaders → SetLocale → auth (session)

|

▼

EnsurePortalAccess middleware (checks 60+ role map)

→ 403 if role not allowed for this portal

|

▼

Portal Controller (MedicalId/ namespace) → Blade View

Request Flow — B2B Connect API

Partner System

1. POST /api/v1/connect/auth/token (client_id + client_secret in body)

2. Bearer token issued with scopes

3. POST /api/v1/connect/patients/search (VerifyIntegrationClient + throttle.client:200,1)

4. GET /api/v1/connect/patients/{id}/summary

→ RequireConsentGrant middleware checks patients:read scope

5. POST /api/v1/connect/records/encounters

→ IdempotencyProtection + consent.grant:patients:write

Request Flow — Mobile Patient

```
Flutter App

1. POST /api/mobile/auth/login-email    (email + password, throttle:5,1)

    or POST /api/mobile/auth/login → /auth/otp/verify (phone+PIN+OTP)

2. Bearer token stored in Flutter Secure Storage

3. GET  /api/mobile/health-id-card      (auth.mobile middleware)

4. GET  /api/mobile/timeline

5. POST /api/mobile/consent-requests/{id}/approve

6. POST /api/mobile/appointments        (book)
```

5. Tech Stack

Layer	Technology	Notes
Backend framework	Laravel 13.8, PHP 8.3+	Monolithic, modular service layer
Database	PostgreSQL	Read replica support, 121 migrations
Cache / Session	Redis	Session driver, cache store
Queue	Redis + Laravel Horizon	Background jobs, queue monitoring dashboard
File storage	Local (Laragon dev) → S3-compatible	Config via <code>config/filesystems.php</code>
PDF generation	barryvdh/laravel-dompdf ^2.2	Documents, reports, exports
QR codes	chillerlan/php-qrcode ^6.0	Health ID QR generation
Backup	spatie/laravel-backup ^10.2	Encrypted automated backups
Web frontend	Blade + Alpine.js / Livewire	Server-rendered portals
Frontend build	Vite + npm	<code>vite.config.js</code>
Mobile app	Flutter 3.3+ (Dart 3.3+)	iOS + Android + Web targets
Mobile state	Riverpod 2.5.1	Feature-based modular providers
Mobile HTTP	Dio 5.4.3	Interceptors, token injection
Mobile routing	GoRouter 13.2.0	Declarative navigation
Mobile storage	Flutter Secure Storage 9.0.0	Tokens never in SharedPreferences
Mobile icons	Lucide Icons ^0.257.0	
Mobile fonts	Google Fonts ^6.2.1	
Mobile shimmer	shimmer ^3.0.0	Loading skeletons

Backend framework	Laravel 13.8, PHP 8.3+	Monolithic, modular service layer
Mobile i18n	intl ^0.19.0	
API standards	REST + FHIR R4	Interoperability layer
HL7 messaging	Custom (<code>config/hl7.php</code>)	Legacy HIS sync via Bridge
Public health	DHIS2 (<code>config/dhis2.php</code>)	Disease surveillance push
KMS	AWS KMS or local (<code>config/kms.php</code>)	Patient PII encryption
USSD	Africa's Talking	Low-tech patient access
WhatsApp	WhatsApp Business API	Notification delivery channel
SMS	Configurable provider	Notification delivery channel
Auth	Custom token per client type	JWT-style, scoped per role
Dev environment	Laragon (Windows)	Virtual host <code>opescare.test</code>
CI/CD	GitHub Actions	<code>deploy.yml</code> , <code>snky-security.yml</code>
Security scanning	Snyk (<code>.snyk</code>)	Dependency vulnerability scanning

6. Subdomain Map

14 production subdomains — all served by the same Laravel app, routed by Nginx:

Subdomain	Purpose
<code>app.opescare.com</code>	Main portal (staff, patient, admin, insurance)
<code>api.opescare.com</code>	REST API for B2B integrations
<code>connect.opescare.com</code>	Connect Platform (B2B interoperability)
<code>fhir.opescare.com</code>	FHIR R4 endpoints
<code>mobile-api.opescare.com</code>	Patient + provider mobile API
<code>lite.opescare.com</code>	OpesCare Lite (low-connectivity)
<code>academy.opescare.com</code>	Training & certification platform
<code>developer.opescare.com</code>	Developer self-service portal
<code>docs.opescare.com</code>	API documentation
<code>caremap.opescare.com</code>	Verified Care Access Map
<code>bridge.opescare.com</code>	Bridge agent sync endpoint
<code>public-health.opescare.com</code>	Public health reporting
<code>ussd.opescare.com</code>	USSD interface (Africa's Talking)
<code>status.opescare.com</code>	System status page

Subdomain enforcement: `EnforceSubdomainScope` middleware + `SUBDOMAIN_ROUTING` env flag.

7. Module Map — 33 Modules Built

Module 1: Patient Identity (Health ID)

Status: Built

- Health ID issuance, QR code generation (`chillerlan/php-qrcode`)
- Temporary QR for facility visits
- Public verification endpoint (`/verify/health-id` , `/verify/qr/{token}`)
- Health ID card endpoint (`GET /mobile/health-id-card`)
- B2B resolution: `POST /api/v1/connect/patients/resolve` (find or auto-create)
- B2B verification: `POST /api/v1/connect/medical-ids/verify` , `verify-qr`
- Duplicate detection and merge cases (`DuplicateMergeController`)
- Master Patient Index (`PatientIdentityService`)
- Controllers: `MedicalId/VerifyController` , `Connect/HealthIdResolutionController` , `Connect/DuplicateMergeController`

Module 2: Authentication & Authorization

Status: Built

Auth type	Endpoint / Mechanism
Web portal login (all roles)	<code>POST /login</code> → session
Patient mobile: email+password	<code>POST /mobile/auth/login-email</code>
Patient mobile: phone+PIN+OTP	<code>POST /mobile/auth/login</code> + <code>/auth/otp/verify</code>
Provider mobile: phone+OTP	<code>POST /provider-mobile/auth/login</code> + <code>/auth/otp/verify</code>
B2B client credentials	<code>POST /api/v1/connect/auth/token</code>
SDK token	<code>VerifySdkToken</code> middleware
Bridge agent	<code>VerifyBridgeAgent</code> middleware
Portal RBAC	<code>EnsurePortalAccess</code> middleware (60+ roles)
Staff invite	<code>GET/POST /invite/{token}</code>
Facility selector	<code>GET/POST /select-facility</code> (multi-facility users)
Language switch	<code>GET /lang/{locale}</code> (en, fr)
OTP challenge	<code>GET/POST /verify/otp</code> + resend
Password reset	<code>/forgot-password</code> , <code>/reset-password/{token}</code>
Demo login-as	<code>POST /demo-access/login-as</code> (demo mode only)

Module 3: Appointments

Status: Built

API (`routes/api.php`):

- `GET/POST /api/v1/appointments` — list, create
- `POST /api/v1/appointments/{id}/reschedule` , `/cancel` , `/check-in`
- `POST /api/v1/appointments/no-shows`

- `GET/POST /mobile/appointments` — patient mobile list + **book**
- `POST /mobile/appointments/{id}/cancel`
- `GET /mobile/facilities/{id}/slots` — available slots

Staff portal (`routes/web.php`):

- Create, confirm, cancel, check-in, no-show
- `AppointmentService` in `app/Modules/Appointments/Services/`

Analytics: `GET /api/v1/analytics/appointments`

Module 4: Queue Management

Status: Built

- `GET /api/v1/queues/tickets` , `POST /queues/check-ins`
 - Call next, start service, transfer, prioritize, complete, cancel
 - `GET /api/v1/queues/display` — public display board
 - Staff portal: check-in, call, start, complete
 - Provider mobile: `GET /provider-mobile/tasks` + call + complete
 - `QueueService` in `app/Modules/Queue/Services/`
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Module 5: Clinical Encounters & Visits (CDSS included)

Status: Built

Visit flow (web portal):

- `POST /portals/staff/visits` → store
- `POST /portals/staff/visits/{id}/transition` → state change
- `POST /portals/staff/visits/{id}/triage` + triage escalate
- `POST /portals/staff/visits/{id}/consult` → consultation notes

Triage API:

- `GET/POST /api/v1/triage` — active triage list, create
- `POST /api/v1/triage/{id}/score` , `/reassess` , `/escalate`

CDSS (Clinical Decision Support System) — Staff Portal:

- `GET /portals/staff/cdss/rules` — clinical rules
- `GET /portals/staff/cdss/lab-rules` — lab threshold rules
- `GET /portals/staff/cdss/drug-interactions` — drug-drug interaction checker
- `GET /portals/staff/cdss/patients/{id}/alerts` — patient-specific alerts
- `GET /portals/staff/cdss/visits/{id}/alerts` — visit-specific alerts
- `POST /portals/staff/cdss/run-checks` — run checks on demand
- `POST /portals/staff/cdss/alerts/{id}/acknowledge` , `/override` , `/dismiss`

Connect API (B2B):

- `POST /api/v1/connect/records/encounters` (consent.grant:patients:write + idempotency)
- `GET /api/v1/connect/patients/{id}/summary` (consent.grant:patients:read)
- `GET /api/v1/connect/patients/{id}/emergency-profile`

Services: `ConsultationService` , `VisitManagementService`

Module 6: Laboratory

Status: Built

- `POST /api/v1/connect/records/lab-results` (B2B write, consent.grant:labs:write)
 - `GET/POST /mobile/labs` , `GET /mobile/labs/{id}` — patient mobile
 - Lab portal: orders, results, samples, `mark-collected` , `mark-processing`
 - Staff portal: `GET /portals/staff/lab-orders`
 - Admin portal: `GET /portals/admin/clinical/lab-orders`
 - FHIR: `GET /fhir/R4/DiagnosticReport` , search + individual
 - LOINC code mapping (admin code system mappings)
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Module 7: Pharmacy & Prescriptions

Status: Built

- `POST /api/v1/connect/records/prescriptions` (B2B write, consent.grant:prescriptions:write)
 - `GET/POST /mobile/prescriptions` , `GET /mobile/prescriptions/{id}` — patient mobile
 - Pharmacy portal: prescriptions, dispense, inventory, controlled substances view
 - Staff portal: `GET /portals/staff/prescriptions` (clinical register)
 - Admin portal: `GET /portals/admin/clinical/prescriptions`
 - Drug formulary: `GET /api/v1/pharmacy/formulary/search` , controlled list, store, toggle availability
 - Controlled substances: dispense, witness confirmation, reconcile, log, inventory
 - FHIR: `GET /fhir/R4/MedicationRequest`
 - ATC code mapping (admin)
 - `DrugFormularyController` , `ControlledSubstanceController`
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Module 8: Billing & Finance

Status: Built

- `GET/POST /api/v1/billing/invoices` — list, create
 - `POST /billing/invoices/{id}/payments` — record payment
 - `POST /billing/payments/{id}/refund`
 - `POST /billing/wallets/deposit`
 - `POST/POST /billing/cashier-sessions` — open, close
 - Payment plans: `POST /api/v1/payment-plans` , `GET /payment-plans/{id}` , installment payment, patient plans list
 - Revenue cycle: summary, aging, denials, trend
 - Insurance FHIR: `GET /fhir/R4/Coverage`
 - Staff portal: billing create, pay
 - `BillingService` , `PaymentService` in `app/Modules/Billing/Services/`
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Module 9: Insurance

Status: Built

API:

- `POST /api/v1/insurance/eligibility/check`
- `POST /api/v1/insurance/preauth` , `/preauth/{id}/decide`
- `GET/POST /api/v1/insurance/claims` , submit, decide, post payment

Insurance portal (web):

- Providers: list, create, add plans
- Policies: list, create, activate, deactivate, add eligibility rules
- Preauths: list, create, submit, decide (approve/deny/partial), cancel
- Claims: list, create, submit, decide, cancel, mark paid

Module 10: Referrals

Status: Built

- `GET/POST /api/v1/referrals` , show, send, accept, reject, complete, cancel
- `POST /api/v1/referrals/expire-stale` — auto-expiry
- Staff portal: list, create, send, accept, reject, complete, cancel (7 actions)
- `ReferralService` in `app/Modules/Referral/Services/`

Module 11: Consent & Governance

Status: Built

- `POST /api/v1/connect/consents/request` (B2B: request consent from patient)
- `POST /api/v1/connect/consents/verify` (check active consent)
- Patient mobile: list, approve, deny, revoke consent
- Patient web portal: approve, deny
- `RequireConsentGrant` middleware enforces on every B2B data read/write
- Consent scopes: `patients:read` , `patients:write` , `labs:write` , `prescriptions:write`
- `ConsentManagementService` + `ConsentController` in Connect namespace
- FHIR: `GET /fhir/R4/Consent`

Patient rights (admin):

- Emergency access: request (B2B), review (admin), audit
- Data correction requests: admin list, approve, reject
- Data export requests: patient create + list + download; admin approve, reject
- Account closures: admin review
- Privacy complaints: admin resolve
- Minor age transitions: admin manage
- Country policies: list, create, update, publish
- Security incidents: list, create, contain, resolve

Services: `AccessLogService` , `CorrectionRequestService` , `DataExportService` , `CountryPolicyService` , `EmergencyAccessService`

Module 12: Verifiable Documents

Status: Built

- `GET/POST /api/v1/documents` — list, create (digitally signed)
- `POST /documents/{id}/amend` , `/revoke` , `/entered-in-error`
- `POST /document-verification/verify-code`
- `POST /documents/{id}/share-links` — time-limited share links
- Public: `GET /verify/document/{token}` , `GET /share/document/{token}`
- Document render: `GET /documents/{id}/view`
- FHIR: `GET /fhir/R4/DocumentReference`

Module 13: Ward & Admissions

Status: Built

- `POST /api/v1/ward/admissions` — admit
- `POST /admissions/{id}/assign-bed` , `/transfer` , `/discharge`
- `POST /admissions/{id}/nursing-round` , `/discharge-plan`
- `GET /api/v1/ward/beds/availability`
- Staff portal: wards list, create ward, admissions, admit, discharge, transfer
- Analytics: `GET /portals/staff/analytics/ward`

Module 14: Maternity & Antenatal

Status: Built

- `GET/POST /api/v1/maternity/patients/{id}/pregnancies`
- `GET /maternity/pregnancies/{id}` , antenatal visits (list+create), deliveries (list+create)
- Models: `Pregnancy` , `AntenatalVisit` , `Delivery`

Module 15: Telemedicine

Status: Built (video provider not yet wired)

- `POST /api/v1/telemedicine/consultations` — book
- `GET /consultations/{id}` , cancel, record consent, join waiting room, initiate call, end call
- Staff portal: list, create, waiting room, call next, show, consent, start, end, cancel
- `TelemedicineController` in both web and API

Gap: Video call provider not integrated. Session model is ready; needs Jitsi, Daily.co, or similar.

Module 16: Public Health Surveillance

Status: Built

Phase 1 (Reports):

- `GET /api/v1/public-health/report-types` , reports list, single report
- `POST /reports/generate-drafts`
- `GET /reports/{id}/quality-checks` , dashboard, facility-dashboard

Phase 2 (Governance):

- Submit for review, assign, approve, request correction, reject, cancel, correct
- `GET /reports/{id}/versions` , status-history, review-queue

Phase 3 (Submissions):

- Submission profiles (list, create)
- Submit report, export, download export, integration status

Phase 4 (Intelligence):

- `GET /api/v1/public-health/signals` , single signal
- `POST /signals/trigger-detection` , `POST /signals/{id}/review`
- Intelligence dashboard, trends, shortages

Services: `DraftGenerationService` , `DataQualityCheckService` , `ExportService` , `SignalDetectionService`

Module 17: Immunizations

Status: Built

- `GET/POST /api/v1/immunizations` — list, record
 - `POST /immunizations/schedule` — create schedule; `GET /immunizations/schedule` — patient schedule
 - `GET /immunizations/{id}` , `POST /{id}/adverse-reactions`
 - Staff portal: list, record form, store
 - Patient mobile: `GET /mobile/immunizations`
 - FHIR: `GET /fhir/R4/Immunization`
 - `ImmunizationService` in `app/Modules/Immunization/Services/`
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Module 18: CareMap (Verified Facility Directory)

Status: Built

Public (no auth):

- `GET /api/v1/care-map/facilities` , show, search, nearby
- `GET /care-map/pharmacies/medicine-search`
- `GET /care-map/labs/test-search`
- `GET /care-map/blood/search`
- `GET /care-map/emergency`
- Web: `GET /care-map` , facility profile, emergency page

Authenticated:

- `POST /care-map/facilities/{id}/save` — save to favourites
- `POST /care-map/facilities/{id}/report` — report inaccuracy
- `POST /care-map/facilities/{id}/claim` — claim facility
- `POST /care-map/partner/facilities/{id}/stock-sync` — partner sync

Admin:

- `POST /admin/care-map/facilities/{id}/verify`
- `POST /admin/care-map/facilities/{id}/suspend`
- `GET /admin/care-map/governance`

Services: `FacilityVerificationService` , `FacilityReportService` , `FacilityFreshnessService` , `InsuranceNetworkSearchService` , `MapProviderService` , `PharmacyStockSearchService` , `BloodAvailabilitySearchService` , `LabTestSearchService`

Module 19: Connect Platform (B2B Interoperability)

Status: Built

- Token issuance (`POST /api/v1/connect/auth/token` , `client_credentials`)
- Widget sessions (`POST /connect/widget/sessions`)
- Patient search (`POST /connect/patients/search`)
- Health ID resolve/verify
- Consent request + verify
- Emergency access request + emergency profile
- Record pulls: patient summary (`patients:read` scope), emergency profile
- Record writes: encounters, lab-results, prescriptions (all require consent + idempotency)
- Inventory sync: pharmacy stock, blood stock

- Webhooks: create subscription, replay event
- Reconciliation cases: list, resolve
- Per-client rate limit: 200 req/min

Separate Controllers in Connect namespace:

- `AuthController` , `PatientSearchController` , `HealthIdResolutionController`
- `ConnectGovernanceController` , `RecordController` , `InventoryController`
- `WebhookController` , `ReconciliationController`
- `MedicalIdVerificationController` , `ConsentController` , `EmergencyAccessController`
- `DuplicateMergeController`

Module 20: FHIR R4 API

Status: Built (read-only + Subscription + Bulk Export)

Resource	Endpoints
CapabilityStatement	<code>GET /fhir/R4/metadata</code> (public)
Patient	search, get, <code>\$everything</code> (consent required)
Encounter	search, get
DiagnosticReport	search, get
MedicationRequest	search, get
Practitioner	search, get
Organization	search, get
DocumentReference	search, get
Consent	search, get
Coverage	search, get
Immunization	search, get
AllergyIntolerance	search, get
Condition	search, get
Subscription	list, create, get, delete
Bulk Export	<code>GET /fhir/R4/\$export</code> , <code>GET /Patient/{id}/\$export</code>

Auth: `VerifyIntegrationClient` on all except metadata.

Gap: FHIR write (POST/PUT individual resources) not implemented.

Module 21: OpesCare Lite (Low-Connectivity)

Status: Built

API:

- Device registration, config sync, push/pull sync
- Offline events, conflict resolution, formulary download

Lite portal (web, simplified UI):

- Dashboard, patient lookup, register patient, check-in
- Consultation, billing, device management (activate/revoke)
- Conflicts list + resolve, offline events per device

`SyncService` in `app/Modules/Offline/Services/`

Module 22: Bridge Agent

Status: Built

- `POST /api/v1/bridge/sync` — sync payload from legacy HIS
 - `POST /bridge/heartbeat` — agent keepalive
 - `GET /bridge/status`
 - Rate limit: 300 req/min, `VerifyBridgeAgent` auth
 - HL7 messaging support (`config/hl7.php`)
 - Admin portal: bridge management (create, toggle, batches)
 - `bridge-agent/config.json` — agent configuration
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Module 23: SDK

Status: Built

- `GET /api/v1/sdk/patients/{id}/summary` (scope: `read_records`)
 - `GET /sdk/patients/{id}/encounters` (scope: `read_records`)
 - `GET /sdk/facilities/{id}` (scope: `read_facility`)
 - `GET /sdk/facilities/{id}/stock` (scope: `read_stock`)
 - `POST /sdk/appointments` (scope: `write_appointments`)
 - `GET /sdk/appointments/{id}` (scope: `read_appointments`)
 - `POST/DELETE /sdk/webhooks/subscriptions` (scope: `manage_webhooks`)
 - `GET /sdk/token/introspect`
 - Rate limit: 120 req/min, `VerifySdkToken` middleware
 - **PHP SDK:** `sdk/php/src/Client.php`
 - **Connect Widget:** `widget/connect-widget.html`
 - **OpenAPI contracts:** `contracts/openapi/opescare-connect-v1.yaml` , `opescare-mobile-v1.yaml`
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Module 24: Developer Portal

Status: Built

- Dashboard, onboarding wizard
 - Apps: list, create, view (credentials shown once, rotate secret)
 - Production access requests: list, create
 - Webhook delivery logs per app
 - API usage analytics
 - Admin side: approve/reject production requests, suspend developer accounts
 - `DeveloperPortalController`
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Module 25: Admin Control Centre

Status: Built

- `/portals/admin/cc/settings` — system settings (get + update)
 - `/portals/admin/cc/feature-flags` — toggle per feature key
 - `/portals/admin/cc/modules` — enable/disable modules
 - `/portals/admin/cc/maintenance` — maintenance windows (create, toggle)
 - `/portals/admin/cc/health` — system health checks
 - `/portals/admin/cc/audit` — audit log viewer
 - Facility go-live readiness: show, create checklist, mark items, approve
 - Facility onboarding: list, show, mark items, approve
 - KPI dashboard: index, trend, export, recompute
 - Code system mappings (LOINC/ICD-10/ATC): list, create, approve, reject, delete
 - Integration certifications: list, create, show, record test run, issue/revoke badge, seed requirements
 - Global search: `GET /api/v1/admin/global-search` (`GlobalSearchService`)
-

Module 26: Communications & Notifications

Status: Built

Routes in `routes/communications.php` :

Notifications:

- List, unread count, mark read, acknowledge, archive

Tasks:

- CRUD, acknowledge, complete, assign, escalate
- `ActionTask` model, `TaskService`

Notification templates (admin):

- CRUD, review workflows, approval, publishing, rollback

Deliveries:

- List, retry failed

Escalation chains:

- CRUD, activate/deactivate
- `AlertEscalationService`

Broadcasts:

- CRUD, publish, cancel
- `BroadcastService`

Messaging threads:

- Threads, messages, participants, assignment, closure
- `MessageThread` , `Message` , `MessageAttachment` , `MessageThreadParticipant`
- `MessagingService` , `MessagePermissionService` , `MessageAttachmentService`

USSD: `POST /ussd/callback` (Africa's Talking, no auth)

Module 27: Security Operations

Status: Built

- `GET /api/v1/security/audit-log` — searchable audit trail
 - `GET /security/suspicious-flags` , review flag
 - `POST /security/breaches` — open breach; notify, close
 - `POST /security/access-reviews` — initiate, complete
 - `POST /security/compliance-exports`
 - Pen test tracker: list tests, open findings, create, show, add finding, update finding
 - Admin portal: incidents (create, update), emergency access review, audit explorer
 - `AddSecurityHeaders` middleware on all responses
-

Module 28: Staff HR & Scheduling

Status: Built

API (`routes/api.php`):

- `GET /api/v1/staff` — directory
- `GET /staff/{id}` , `PATCH /staff/{id}` — profile
- `GET /staff/rosters` , `POST /staff/shifts` , `DELETE /staff/shifts/{id}`
- `POST /staff/leave` — request, approve, reject

Staff portal (`routes/web.php`):

- **Directory:** list, create, status toggle, add licence
 - **Shifts:** list, create, toggle (enable/disable)
 - **Roster:** list, create, publish, archive, assign staff, unassign
 - **Leave:** list, create, approve, reject, withdraw
-

Module 29: Analytics & Reporting

Status: Built

- `GET /api/v1/analytics/facilities/{id}/dashboard`
 - `GET /analytics/appointments` , queue, billing stats
 - `POST /analytics/exports` , status check
 - Provider performance: summary, top diagnoses, facility summary
 - Revenue cycle: summary, aging, denials, trend
 - Survey reports: `GET /api/v1/reports/surveys/satisfaction`
 - Staff portal analytics: queue, ward, financial, data-quality dashboards
 - `AnalyticsDashboardController` (web)
-

Module 30: Care Plans & Advance Directives

Status: Built

- `POST /api/v1/care-plans` , `GET /care-plans/{id}`
 - `POST /care-plans/{id}/goals` , `PATCH /goals/{goalId}`
 - `POST /care-plans/{id}/interventions`
 - Patient mobile (read-only): `GET /mobile/care-plans` , `GET /mobile/care-plans/{id}`
 - Advance directives: `GET/POST /api/v1/patients/{id}/advance-directives` , show, destroy
-

Module 31: Academy (Training & Competency)

Status: Built

Routes in `routes/academy.php` :

- Courses (list, enroll), lessons, completion
- Quizzes: start, submit
- Simulations: start, submit
- Certificates: verify (public `/verify/certificate/{token}`), revoke, renew
- Trainer sign-offs
- Facility readiness dashboard
- Academy learner dashboard (auth web)

Services: `EnrollmentService` , `QuizService` , `SimulationService` , `CertificateService` , `CertificateVerificationService` , `CompetencyGateService` , `AcademyReportingService` , `CourseService`

Module 32: Partner Management

Status: Built

Routes in `routes/partners.php` :

- Partner governance: list, approve, suspend
- Partner documents verification
- Agreements + certifications
- Access + contribution permissions
- Integration cases
- Partner dashboard

Partner module has rich modelling:

- `Partner` , `PartnerFacility` , `PartnerProfessional` , `PartnerContact`
- `PartnerAgreement` , `PartnerDocument` , `PartnerIntegration` , `PartnerGovernanceCase`
- `PartnerAccessPermission` , `PartnerContributionPermission` , `PartnerContribution`
- `PartnerQualityScore` , `PartnerRiskScore` , `PartnerAuditLog`
- Enums: `PartnerType` , `PartnerStatus` , `TrustLevel`
- Policies: `PartnerPolicy` , `PartnerDocumentPolicy` , `PartnerAgreementPolicy` , `PartnerPermissionPolicy` , `PartnerGovernanceCasePolicy` , `PartnerIntegrationPolicy`
- Services: `PartnerApplicationService` , `PartnerPermissionService` , `PartnerAuditService` , `PartnerVerificationService` , `PartnerAgreementService` , `PartnerContributionService` , `PartnerQualityScoreService` , `PartnerRiskScoreService` , `PartnerIntegrationGovernanceService`

Module 33: Subscriptions & SaaS Billing

Status: Built (manual, not automated)

- `GET /api/v1/subscriptions/plans` , plan detail
- `GET /subscriptions/my` — current subscription
- `POST /subscriptions` — subscribe, upgrade, cancel
- `GET /subscriptions/usage` , feature limit check
- Admin portal: plan CRUD + toggle, subscription management (detail, cancel, renew, pause, reactivate, change plan), invoices + mark-paid

Gap: No automated payment gateway (Paystack/Stripe). Billing is currently manual admin action.

Module 34: Data Import

Status: Built

Staff portal data import flow (upload → map → validate → preview → approve → rollback):

- Upload file (`POST /portals/staff/data-import`)
 - Column mapping (`GET/POST /data-import/{id}/mapping`)
 - Validate (`POST /data-import/{id}/validate`)
 - Preview (`GET /data-import/{id}/preview`)
 - Approve (`POST /data-import/{id}/approve`)
 - Rollback (`POST /data-import/{id}/rollback`)
 - Cancel + audit log
-

Module 35: Supply Chain

Status: Built

Staff portal supply chain:

- Items: list, create
 - Suppliers: list, create
 - Stock: list, receive stock, adjust
 - Purchase orders: list, create, approve
 - Goods receipts: list, create
 - Stock movements: list
-

Module 36: Radiology

Status: Built (API only — no portal UI)

- `POST /api/v1/radiology/reports` — store report
- `GET /radiology/reports/{id}` , finalize, amend, distribute
- `GET /radiology/facilities/{id}/reports/pending`

Gap: No staff portal view for radiology results. API is complete.

Module 37: File Storage

Status: Built

Staff portal file management:

- `GET /portals/staff/files` — list files
 - `GET /files/upload` → `POST /files` — upload attachment
 - `GET /files/{id}/download`
 - `DELETE /files/{id}`
-

Module 38: Patient Satisfaction Surveys

Status: Built

- `GET /api/v1/reports/surveys/satisfaction` — reporting endpoint
- Patient mobile: `GET /mobile/surveys` , show, `POST /surveys/{id}/submit`

- `MobileSurveyController`

Module 39: Medical Record Export

Status: Built

Patient mobile:

- `POST /mobile/medical-records/export/pdf` — PDF export
- `POST /mobile/medical-records/export/fhir` — FHIR bundle export
- `MedicalRecordExportController`

Module 40: Demo Mode

Status: Built

- `GET /demo-access` — demo landing
- `GET /demo-access/public` , `/demo-access/internal`
- `POST /demo-access/login-as` — login as any demo persona
- `POST /api/demo/reset` — reset demo data (calls artisan command)
- `DemoAccessController` , `DemoDataScope` middleware, `DemoSessionMiddleware`
- `config/demo.php` , seeder: `DemoSeedCommand` , reset: `DemoResetCommand`

8. Service Layer — `app/Modules/`

The service layer lives in `app/Modules/`. Each module follows the pattern:

```
app/Modules/{Module}/
  Services/      - Business logic (called from controllers)

  Models/       - Eloquent models specific to this module

  Policies/     - Laravel authorization policies

  Enums/        - PHP 8.1+ enums
```

Confirmed modules with service classes:

Module	Key Services
<code>AccessControl</code>	<code>EmergencyAccessService</code>
<code>Appointments</code>	<code>AppointmentService</code>
<code>Academy</code>	<code>EnrollmentService</code> , <code>QuizService</code> , <code>SimulationService</code> , <code>CertificateService</code> , <code>CertificateVerificationService</code> , <code>CompetencyGateService</code> , <code>AcademyReportingService</code> , <code>CourseService</code>
<code>Billing</code>	<code>BillingService</code> , <code>PaymentService</code>
<code>Broadcasts</code>	<code>BroadcastService</code>

AccessControl	EmergencyAccessService
CareMap	FacilityVerificationService , FacilityReportService , FacilityFreshnessService , InsuranceNetworkSearchService , MapProviderService , PharmacyStockSearchService , BloodAvailabilitySearchService , LabTestSearchService
ConsentManagement	ConsentManagementService
EncounterManagement	ConsultationService , VisitManagementService
FacilityReadiness	FacilityGoLiveService
Governance	AccessLogService , CorrectionRequestService , DataExportService , CountryPolicyService , EmergencyAccessService
Immunization	ImmunizationService
Messaging	MessagingService , MessagePermissionService , MessageAttachmentService
Notifications	NotificationService , NotificationTemplateRenderer , NotificationPreferenceService , AlertEscalationService , VoiceNotificationService
Offline	SyncService
OperationalFlow	PatientJourneyService
Partners	PartnerApplicationService , PartnerPermissionService , PartnerAuditService , PartnerVerificationService , PartnerAgreementService , PartnerContributionService , PartnerQualityScoreService , PartnerRiskScoreService , PartnerIntegrationGovernanceService
PatientIdentity	PatientIdentityService
PublicHealth	DraftGenerationService , DataQualityCheckService , ExportService , SignalDetectionService
Queue	QueueService
Referral	ReferralService
Search	GlobalSearchService
Support	SupportService
Tasks	TaskService

9. Data Model Overview

Core identity cluster

```
User (auth) — Patient — HealthId
    |
    |— Guardian (linked patients)
    |— ConsentGrant (per scope, per client)
    |— ConsentAudit (automatic on each access)
    |— AccessLog (every data touch)
    |— FamilyLink
```

Clinical cluster

Patient

- └ Appointment (AppointmentSlot, AppointmentType, AppointmentWaitlist)

- └ Queue (QueueTicket, QueueDisplay)

- └ Visit / Encounter

 - └ Diagnosis (ICD-10)

 - └ Procedure

 - └ VitalSign

 - └ Referral

- └ LabOrder → LabResult

- └ Prescription (MedicationRequest)

- └ AllergyRecord

- └ ConditionRecord

- └ ImmunizationRecord

- └ AntenatalVisit, Delivery, Pregnancy

- └ Admission → Bed, BedAssignment, NursingRound, DischargePlan

- └ CarePlan → CareGoal, CareIntervention

- └ AdvanceDirective

Governance cluster

ConsentGrant — ConsentAudit

ConsentRequest (pending approval)

EmergencyAccessRequest — EmergencyAccessEvent

CorrectionRequest

DataExportRequest

AccountClosureRequest

PrivacyComplaint

MinorTransition

CountryPolicy

SecurityIncident, BreachEvent

AccessReview, AuditEvent, AuditExport

Integration cluster

IntegrationClient (B2B credential)

└─ ApiScopeGrant

└─ Webhook — WebhookDelivery — WebhookEvent

└─ HealthIdResolution, HealthIdQrToken

Partner — PartnerFacility, PartnerProfessional, PartnerContact

└─ PartnerAgreement, PartnerDocument, PartnerIntegration

└─ PartnerGovernanceCase, PartnerAccessPermission

└─ PartnerContributionPermission, PartnerContribution

└─ PartnerQualityScore, PartnerRiskScore, PartnerAuditLog

LiteDevice — OfflineSync — OfflineEvent

BridgeAgent — BridgeBatch

Notifications/Comms cluster

NotificationTemplate — NotificationEvent — NotificationDelivery

NotificationPreference

EscalationChain — EscalationRule

VoiceNotificationJob

ActionTask

MessageThread — Message — MessageAttachment

MessageThreadParticipant

Broadcast

Operational cluster



Financial cluster



Analytics cluster



10. Portal System & RBAC

Portal routing

The `EnsurePortalAccess` middleware reads the authenticated user's role and checks the allowed-roles list per portal. 403 if mismatch.

Portal URL prefix	Roles (60+ total)
<code>/portals/patient</code>	<code>patient</code> , <code>guardian</code> , <code>caregiver</code> , <code>dependent_manager</code> , <code>emergency_contact</code>
<code>/portals/staff</code>	All clinical + operational roles (30+)
<code>/portals/insurance</code>	<code>insurance_*</code> roles
<code>/portals/admin</code>	<code>platform_admin</code> , <code>super_admin</code> , <code>privacy_officer</code> , <code>security_officer</code> , etc.
<code>/portals/developer</code>	<code>developer</code> , <code>developer_org_admin</code> , <code>api_partner</code> , <code>sandbox_developer</code> , etc.
<code>/portals/pharmacy</code>	<code>pharmacist</code> , <code>pharmacy_manager</code> , <code>pharmacy_technician</code> , etc.

/portals/patient	patient, guardian, caregiver, dependent_manager, emergency_contact
/portals/lab	labtech, lab_scientist, lab_manager, lab_validator, sample_collection
/portals/healthorg	ngo_admin, health_program_manager, outreach_team, mobile_clinic_team
/portals/lite	lite_facility, lite_staff, lite_device, lite_offline_sync

Additional middleware per route group

- guardian.context — guardian viewing dependent records
- facility.context — require active facility selection
- api.admin — require admin role for CareMap admin actions
- auth:sanctum — Sanctum auth for mobile care plans, surveys, export

Sidebar partials (role-specific navigation)

resources/views/partials/sidebars/ :

doctor, nurse, pharmacist, pharmacy_manager, lab_manager, labtech, medicine_stock, ngo_health_org, outreach_mobile, sample_collection, specialist_doctor, insurance_admin

11. Integration Surface

Inbound (what connects to OpesCare)

System	Protocol	Auth	Purpose
Partner HIS	REST	client_credentials	Read/write patient records
SDK consumers	REST	SDK token (scoped)	Embed health data
Bridge Agents	REST	VerifyBridgeAgent	Sync from legacy HIS
Africa's Talking	HTTP webhook	none (USSD)	USSD patient interface
OpesCare Lite devices	REST	VerifyIntegrationClient	Offline-capable field devices
Mobile app (patient)	REST	auth.mobile JWT	Patient B2C
Provider mobile app	REST	VerifyIntegrationClient	Clinical staff mobile

Outbound (what OpesCare connects to)

System	Purpose	Config
DHIS2	Monthly public health push	config/dhis2.php
HL7	Legacy HIS bridge messaging	config/hl7.php
WhatsApp Business API	Notification delivery	WHATSAPP_PHONE_NUMBER_ID, WHATSAPP_ACCESS_TOKEN
SMS provider	Notification delivery	config/services.php
Email (SMTP)	Notification delivery	config/mail.php
Push (FCM/APNs)	Mobile push notifications	PushProvider

DHIS2	Monthly public health push	<code>config/dhis2.php</code>
Voice	Voice call notifications	<code>VoiceNotificationService</code>
AWS KMS	Encryption key management	<code>config/kms.php</code>
S3-compatible	File storage (prod)	<code>config/filesystems.php</code>
Redis / Horizon	Queue + monitoring	<code>config/horizon.php</code>

12. Notification System — 5 Channels

The notifications module supports 5 delivery channels via dedicated provider classes in `app/Modules/Notifications/Providers/`:

Channel	Provider class	Notes
Email	<code>EmailProvider.php</code>	Laravel Mail/SMTP
WhatsApp	<code>WhatsAppProvider.php</code>	WhatsApp Business API
SMS	<code>SmsProvider.php</code>	Configurable carrier
Push	<code>PushProvider.php</code>	FCM (Android) + APNs (iOS)
Voice	<code>VoiceProvider.php</code> + <code>VoiceNotificationService.php</code>	Voice call delivery

Additional notification infrastructure:

- `NotificationTemplate` + `NotificationTemplateRenderer` — template engine with approval workflow
- `NotificationPreference` + `NotificationPreferenceService` — per-user channel preferences
- `NotificationDelivery` — delivery tracking + retry
- `EscalationChain` + `AlertEscalationService` — escalation rules
- `VoiceNotificationJob` — async voice call queuing

13. Mobile App — Patient

Location: `apps/mobile-patient`

Architecture

Clean Architecture, feature-based modular structure. Each feature:

```
lib/features/{feature}/
  data/           - API calls, local storage, repositories

  models/         - Data models

  presentation/   - Screens, widgets

  providers/      - Riverpod state providers
```

Screen map (15 screens across 11 features)

```

/ (root)
├─ /login          - email+password or phone+PIN+OTP

├─ /otp           - OTP verification step

└─ /home (shell - bottom nav)
    │
    ├─ /home        - Dashboard / timeline feed
    │
    ├─ /health-id   - Digital Health ID card + QR
    │
    ├─ /appointments
    │   │
    │   ├─ /appointments - List (upcoming + past)
    │   │
    │   └─ /appointments/:id - Detail
    │
    ├─ /labs
    │   │
    │   ├─ /labs        - List
    │   │
    │   └─ /labs/:id    - Detail (results)
    │
    ├─ /prescriptions
    │   │
    │   ├─ /prescriptions - List
    │   │
    │   └─ /prescriptions/:id - Detail
    │
    ├─ /consent      - Pending consent requests
    │
    ├─ /documents    - Official documents
    │
    ├─ /access-logs  - Who accessed my data
    │
    ├─ /timeline     - Clinical event history
    │
    └─ /settings     - User preferences + push tokens
  
```

API calls by screen

Screen	Endpoint
Login (email)	POST /mobile/auth/login-email
Login (phone)	POST /mobile/auth/login

Login (email)	POST /mobile/auth/login-email
OTP verify	POST /mobile/auth/otp/verify
Home	GET /mobile/timeline
Health ID	GET /mobile/health-id-card
Appointments	GET /mobile/appointments
Appointment detail	GET /mobile/appointments/{id}
Book appointment	POST /mobile/appointments
Cancel appointment	POST /mobile/appointments/{id}/cancel
Facilities/slots	GET /mobile/facilities , GET /mobile/facilities/{id}/slots
Labs	GET /mobile/labs
Lab detail	GET /mobile/labs/{id}
Prescriptions	GET /mobile/prescriptions
Prescription detail	GET /mobile/prescriptions/{id}
Consent	GET /mobile/consent-requests
Approve consent	POST /mobile/consent-requests/{id}/approve
Deny consent	POST /mobile/consent-requests/{id}/deny
Revoke consent	POST /mobile/consents/{id}/revoke
Access Logs	GET /mobile/access-logs
Documents	GET /mobile/documents
Allergies	GET /mobile/allergies
Clinical	GET /mobile/clinical
Immunizations	GET /mobile/immunizations
Care plans	GET /mobile/care-plans , GET /mobile/care-plans/{id}
Surveys	GET /mobile/surveys , submit
Medical record export	POST /mobile/medical-records/export/pdf , /export/fhir
Settings	GET /mobile/settings , PATCH /mobile/settings
Push token	POST /mobile/push-tokens , DELETE /mobile/push-tokens/{id}
Correction request	POST /mobile/correction-requests
Data export	POST /mobile/data-export-requests , list, download
Offline policy	POST /mobile/offline/policies

14. Artisan Commands & Scheduled Jobs

11 Artisan commands in `app/Console/Commands/` :

Command	Schedule	Purpose
---------	----------	---------

opescare:demo:seed	Manual	Seed demo data for all personas
opescare:demo:reset	Manual / API	Reset demo environment
opescare:check-age-transitions	Daily	Process guardian transitions when minors turn 18
opescare:encrypt-patient-pii	Manual (migration)	Encrypt existing patient PII fields
opescare:import-facility-registry	Manual	Bulk-import facility registry
opescare:import-insurance-registry	Manual	Bulk-import insurance registry
opescare:rotate-secrets	Manual / periodic	Rotate integration client secrets
opescare:push-dhis2-report	1st of month 04:00	Push public health data to DHIS2
opescare:notify-expiring-credentials	Monday 08:00 weekly	Alert about expiring provider credentials
opescare:purge-expired-data	Daily 03:00	Data retention enforcement
opescare:audit-facility-scope	Manual	Audit data that crossed facility boundaries

Service Providers:

- AppServiceProvider
- HorizonServiceProvider — Redis queue monitoring
- ProductionSafetyServiceProvider — guards against dangerous operations in production

15. Contracts, SDK & Widget

OpenAPI Contracts

contracts/openapi/ :

- opescare-connect-v1.yaml — Full OpenAPI spec for the Connect B2B API
- opescare-mobile-v1.yaml — Full OpenAPI spec for the Mobile Patient API

PHP SDK

sdk/php/src/Client.php — PHP client library for consuming the OpesCare API

Connect Widget

widget/connect-widget.html — Embeddable HTML widget for third-party systems to access OpesCare patient data (with consent flow) inside their own interface

Bridge Agent

bridge-agent/config.json — Configuration for the OpesCare Bridge Agent that syncs data from legacy HIS systems

16. CI/CD & Security Scanning

.github/workflows/ :

- deploy.yaml — Automated deployment pipeline
- snyk-security.yaml — Snyk dependency vulnerability scanning

.snyk — Snyk configuration at repo root

Developer scripts (composer.json):

- `composer setup` — Full install, key generate, migrate, npm build
- `composer dev` — Concurrent dev server, queue worker, log viewer, Vite (via `concurrently`)
- `composer test` — Clear config + run PHPUnit

Testing:

- `tests/Feature/` — Academy, Communications, Documents, MedicalId, Partners, SecurityOperations
 - `tests/Unit/` — Unit tests
 - PHPUnit ^12.5.12
-

17. Key Flows

Flow 1: New Patient Registration

1. Patient visits `/signup/patient`
 1. Submits name, phone, email, DOB
 2. OTP sent → `/verify/otp`
 3. Account created → Health ID auto-issued
 4. Mobile app: login → Health ID card available
 5. QR generated → can be scanned at any facility

Flow 2: Complete Facility Visit



Flow 3: B2B Connect Data Access

Partner System

1. POST /api/v1/connect/auth/token → scoped access_token
2. POST /connect/consents/request → consent_request_id
- (Patient mobile approves via /mobile/consent-requests/{id}/approve)
3. GET /connect/patients/{id}/summary → RequireConsentGrant passes
→ ConsentAudit record created automatically
4. POST /connect/records/encounters → IdempotencyProtection key required
5. Webhook fires → partner notified of subsequent updates

Flow 4: Insurance Claim Lifecycle

1. Facility creates invoice for visit
 1. POST /api/v1/insurance/preauth → preauth requested
 2. Insurer decides (approve/partial/deny) via insurance portal
 3. POST /api/v1/insurance/claims → claim submitted
 4. Insurer adjudicates via insurance portal
 5. POST /insurance/claims/{id}/payment → payment posted
 6. Invoice marked paid / partially paid
 7. Revenue cycle report updated

Flow 5: Patient Consent Flow

B2B Partner

1. POST /connect/consents/request → {consent_request_id, scope: "patients:read"}

Patient Mobile App

2. GET /mobile/consent-requests → sees pending request with facility + scope

3. POST /consent-requests/{id}/approve

→ ConsentGrant created with expiry

→ ConsentAudit logged

B2B Partner

4. GET /connect/patients/{id}/summary → RequireConsentGrant passes

5. ConsentAudit entry written on every read

6. Patient can see this in /mobile/access-logs

Flow 6: Public Health Report Submission

Data Steward (Facility)

Phase 1: POST /public-health/reports/generate-drafts

→ DraftGenerationService builds auto-draft from clinical data

GET /reports/{id}/quality-checks

→ DataQualityCheckService runs completeness + consistency checks

Phase 2: POST /reports/{id}/submit-for-review → review queue

Reviewer: approve / request-correction / reject

Phase 3: POST /reports/{id}/submit → goes to submission profiles

→ ExportService generates PDF/Excel

→ POST /reports/{id}/export

Phase 4: PushDhis2ReportCommand fires (1st of month 04:00)

→ sends to DHIS2

Intelligence Layer (automated)

→ SignalDetectionService checks for outbreak signals

→ OutbreakAlert created if threshold exceeded

→ POST /signals/trigger-detection can also be called manually

Flow 7: Emergency Access

Emergency Clinician

1. POST /connect/emergency-access/request → logged immediately
2. GET /connect/patients/{id}/emergency-profile → critical data (auto-granted)

Privacy Officer (Admin)

3. Sees entry in /portals/admin/security/emergency-access
4. Flags if inappropriate → SecurityIncident created
5. POST /security/suspicious-flags/{id}/review

Patient Mobile

6. Sees emergency access in /mobile/access-logs
7. Can file correction request if unauthorized

Flow 8: USSD Patient Access

Patient phones *XXX# (Africa's Talking)

- POST /api/ussd/callback (no auth)
- UssdController processes session state
- Menu: [1] My Health ID [2] Nearest facility [3] Emergency
- Response sent back to Africa's Talking → displayed on phone

18. What's Built vs What's Missing

Confirmed as fully built

Module	Backend API		Web Portal	Mobile
Patient Identity / Health ID				
Auth (all types)				
Appointments (incl. booking)			 booking + cancel	

Patient Identity / Health ID	✓	✓	✓
Queue management	✓	✓	✓ (provider mobile)
Clinical Encounters	✓	✓	read-only
CDSS	—	✓	—
Laboratory	✓	✓ (lab portal)	✓
Pharmacy & Prescriptions	✓	✓ (pharmacy portal)	✓
Billing & Finance	✓	✓	—
Insurance	✓	✓ (insurance portal)	—
Referrals	✓	✓	—
Consent & Governance	✓	✓	✓
Verifiable Documents	✓	✓	✓
Ward & Admissions	✓	✓	—
Maternity & Antenatal	✓	—	—
Telemedicine (stub)	✓	✓	—
Public Health Surveillance	✓	—	—
Immunizations	✓	✓	✓
CareMap	✓	✓	—
Connect Platform (B2B)	✓	✓ (admin)	—
FHIR R4	✓	—	—
OpesCare Lite	✓	✓ (lite portal)	—
Bridge Agent	✓	✓ (admin)	—
SDK	✓	—	—
Developer Portal	—	✓	—
Admin Control Centre	—	✓	—
Communications / Notifications	✓	✓	push tokens
Security Operations	✓	✓	—
HR & Scheduling	✓	✓	—
Analytics & Reports	✓	✓	—
Care Plans	✓	—	✓ read-only
Advance Directives	✓	—	—
Academy	✓	✓	—
Partner Management	✓	—	—
Subscriptions	✓	✓ (admin)	—
Data Import	—	✓	—
Supply Chain	✓	✓	—
Radiology	✓	✗ no portal UI	—

Patient Identity / Health ID	✓	✓	✓
File Storage	—	✓	—
Patient Surveys	✓	—	✓
Medical Record Export	✓	—	✓
Demo Mode	✓	✓	—

Confirmed gaps

Gap	Impact	Effort
Telemedicine video provider not wired (Jitsi/Daily/Twilio)	Telemedicine is a UI stub	Medium
Provider mobile app not scaffolded	No mobile tool for clinical staff	Large
Radiology portal UI (no blade views)	Radiologists have no web interface	Small
Advance directives patient view (mobile)	Patients can't see their DNR	Small
Care plan goal actions (patient)	Care plans are read-only on mobile	Small
Patient portal web — incomplete pages	Non-smartphone patients excluded	Medium
FHIR write (POST/PUT resources)	One-way interoperability only	Large
Subscription auto-billing (Paystack/Stripe)	Manual admin action required	Medium
USSD full session tree	Webhook exists, no menu state	Medium
Mobile offline mode (SQLite + local sync)	No data when no internet	Large
Biometric auth on mobile	Security UX gap	Small
WhatsApp / Push end-to-end test	Unclear if delivery pipeline working	Small
French translations complete	Only English visibly in use	Medium
Data retention scheduled jobs (verify running)	Legal compliance risk	Small
KMS key rotation verified	Security risk if not configured	Small

19. Next Priorities

P0 — Before any go-live (production blockers)

1. **Verify data retention job** — `PurgeExpiredDataCommand` must be scheduled and running. Check `routes/console.php`.
2. **KMS configuration** — confirm `config/kms.php` points to real key management, not a dev fallback.
3. **Auth hardening** — audit token expiry, refresh flows, and revocation for all 4 auth types (web, mobile patient, provider mobile, B2B).
4. **Push notification pipeline** — confirm FCM/APNs tokens flow from `POST /mobile/push-tokens` through to `PushProvider` delivery. Test end-to-end.
5. **WhatsApp delivery** — confirm `WHATSAPP_PHONE_NUMBER_ID` + `WHATSAPP_ACCESS_TOKEN` env vars are set and `WhatsAppProvider` sends real messages.
6. **CORS** — verify `config/cors.php` allowed origins match production domain list, not wildcards.
7. **ProductionSafetyServiceProvider** — review what guards it enforces, ensure they're all active.

P1 — High value, clear path

1. **Provider mobile app** — The entire `Api/ProviderMobile/` API is built. Scaffold a new Flutter app using `apps/mobile-patient` as the template. Screens needed: login, facility select, queue (call + complete), patient scan, patient clinical profile.
2. **Telemedicine video integration** — Pick Jitsi Meet (free, self-host) or Daily.co. The session model (`TelemedicineController`) is complete; wire in the room URL.
3. **Radiology portal UI** — Create blade views under `resources/views/portals/staff/radiology/`. The API (`RadiologyReportController`) is 100% done.
4. **Web patient portal** — Build out pages for patients who don't have smartphones (appointment booking, consent approval, document download, access logs).
5. **Subscription auto-billing** — Integrate Pystack or Stripe. The `SubscriptionController` is built; wire in the payment provider.

P2 — Important, more complex

1. **Mobile offline mode** — Add local SQLite with `sqflite` Flutter package. Use existing `/mobile/offline/policies` + `/offline/policies/{id}/queue` API.
2. **FHIR write operations** — Add POST/PUT on Patient, Encounter, Observation. Needed for government interoperability.
3. **USSD session tree** — Build full menu state machine in `UssdController` : patient lookup by phone, Health ID display, nearest facility, appointment status.
4. **French translations** — Wrap all blade views in `__('...')`, create `lang/fr/` files.

P3 — Quality & completeness

1. **Advance directives patient view** — Add screen in mobile app, call `GET /mobile/` directives endpoint.
2. **Care plan goal check-off** — Allow patients to mark goals complete from mobile.
3. **Biometric auth** — Add `local_auth` Flutter package to `apps/mobile-patient`.
4. **Survey prompt UX** — Auto-show survey after appointment completion in mobile.
5. **Multi-facility provider switching** — The API supports `POST /provider-mobile/facilities/{id}/switch`. Wire it in the future provider app's facility selector.
6. **Academy mobile access** — Add course catalogue + quiz screens to patient (or provider) app.
7. **Blood bank portal UI** — The staff portal blood inventory is under inventory; consider a dedicated blood bank view.

20. Key Files Reference

Path	Purpose
<code>apps/api-laravel/routes/api.php</code>	All REST API routes (909 lines)
<code>apps/api-laravel/routes/web.php</code>	All web portal routes (641 lines)
<code>apps/api-laravel/routes/communications.php</code>	Notifications, tasks, messaging
<code>apps/api-laravel/routes/academy.php</code>	Training routes
<code>apps/api-laravel/routes/partners.php</code>	Partner governance routes
<code>apps/api-laravel/app/Http/Middleware/EnsurePortalAccess.php</code>	RBAC enforcement
<code>apps/api-laravel/app/Http/Middleware/RequireConsentGrant.php</code>	Consent scope enforcement
<code>apps/api-laravel/app/Http/Middleware/IdempotencyProtection.php</code>	Idempotency for writes
<code>apps/api-laravel/config/opescare.php</code>	System-wide OpesCare config
<code>apps/api-laravel/config/screen_registry.php</code>	Portal screen/route mapping

apps/api-laravel/routes/api.php	All REST API routes (909 lines)
apps/api-laravel/config/dhis2.php	DHIS2 integration config
apps/api-laravel/config/hl7.php	HL7 messaging config
apps/api-laravel/config/kms.php	Key management config
apps/api-laravel/config/demo.php	Demo mode config
apps/api-laravel/config/data_retention.php	Data retention policies
apps/api-laravel/app/Modules/	Service layer (40+ modules)
apps/api-laravel/app/Console/Commands/	11 Artisan commands
apps/mobile-patient/lib/	Flutter patient app
apps/mobile-patient/lib/core/router/	Navigation config
apps/mobile-patient/pubspec.yaml	Flutter dependencies
contracts/openapi/opescare-connect-v1.yaml	Connect API OpenAPI spec
contracts/openapi/opescare-mobile-v1.yaml	Mobile API OpenAPI spec
sdk/php/src/Client.php	PHP SDK client
widget/connect-widget.html	Embeddable B2B widget
bridge-agent/config.json	Bridge agent config
DEPLOYMENT.md	Production deployment guide
ONBOARDING.md	Developer onboarding
PROJECT_KNOWLEDGE.md	Project knowledge base
.github/workflows/deploy.yml	CI/CD pipeline
.github/workflows/snyk-security.yml	Security scanning

For individual feature specs, use `/spec` to file a GitHub issue per item above. This document covers everything currently in the codebase — nothing omitted.